

Chemistry

CHEMICAL EQUILIBRIUM

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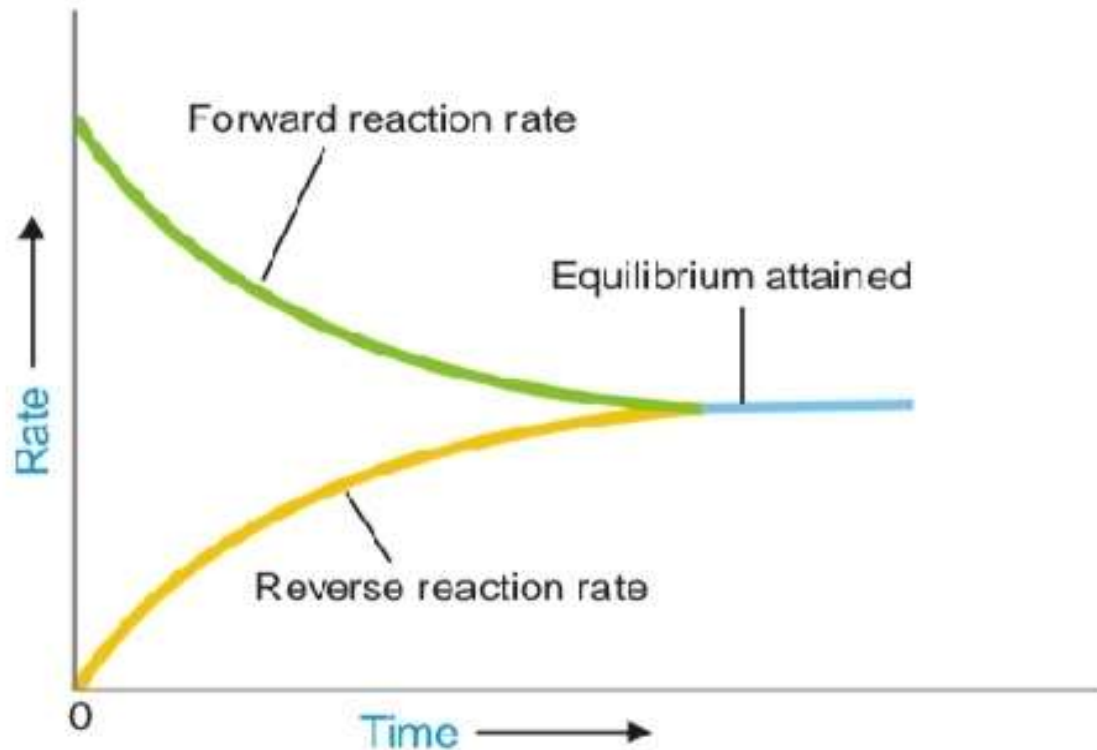
Reversible Reaction

A reaction which can go in the forward and backward direction simultaneously is called a Reversible reaction.



Chemical Equilibrium

The state of a reversible reaction when the two opposing reactions occur at the same rate and the concentrations of reactants and products do not change with time.



At equilibrium the forward reaction rate equals the reverse reaction rate.

Characteristics Of Chemical Equilibrium

- ▶ At equilibrium ,the concentration of each of the reactants and products becomes constant.
- ▶ Rate of forward reaction becomes equal to backward reaction,i.e.equilibrium is dynamic in a nature.
- ▶ A equilibrium is established only if system is closed one.
- ▶ Chemical equilibrium can be attained from both the directions($\rightleftharpoons$) of nature.
- ▶ It is not affected if any catalyst is used..
- ▶ At equilibrium the Gibbs free energy (G) is minimum and any change taking place at equilibrium proceeds without change in free energy i.e. $\Delta G = 0$

LAW OF MASS ACTION

The rate of a chemical reaction is proportional to the active masses of the reactants.

Let us consider a general reaction $A + B \rightleftharpoons C + D$

and let $[A]$, $[B]$, $[C]$ and $[D]$ represent the molar concentrations of A, B, C and D at the equilibrium point. According to the Law of Mass action.

Rate of forward reaction $\propto [A][B] = k_1 [A][B]$

Rate of reverse reaction $\propto [C][D] = k_2 [C][D]$

where k_1 and k_2 are rate constants for the forward and reverse reactions.

At equilibrium, rate of forward reaction = rate of reverse reaction.

Therefore,

$$k_1[A][B] = k_2[C][D]$$

$$\frac{k_1}{k_2} = \frac{[C][D]}{[A][B]}$$

At any specific temperature k_1/k_2 is constant since both k_1 and k_2 are constants. The ratio k_1/k_2 is called Equilibrium constant and is represented by the symbol K_c , or simply k . The subscript 'c' indicates that the value is in terms of concentrations of reactants and products

Equilibrium constant $k_c = \frac{[C][D]}{[A][B]}$

— Product concentrations

— Reactant concentrations

The Equilibrium Constant in Terms Of Partial Pressures

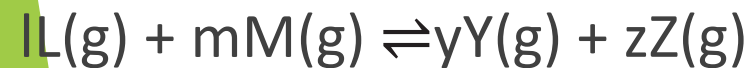
When all the reactants and products are gases, we can also formulate the equilibrium constant expression in terms of partial pressure. The relationship between the partial pressure (p) of any one gas in the equilibrium mixture and the molar concentration follows from the general ideal gas equation

$$pV = nRT \text{ Or, } p = (n/V) RT$$

The quantity (n/V) is the number of moles of the gas per unit volume and is simply the molar concentration. Thus,

$$p = (\text{Molar concentration}) \times RT$$

i.e., the partial pressure of a gas in the equilibrium mixture is directly proportional to its molar concentration at a given temperature



$$K_p = \frac{(p_Y)^y (p_Z)^z}{(p_L)^l (p_M)^m}$$

Relation Between Kc and Kp

Let us consider a general reaction



$$K_p = \frac{(p_C)^l (p_D)^m}{(p_A)^j (p_B)^k}$$

Assuming that all these gases constituting the equilibrium mixture obey the ideal gas equation, the partial pressure (p) of a gas is

$$p = (n/V)RT$$

Where (n/V) is the molar concentration. Thus the partial pressures of individual gases, A, B, C and D are:

$$p_A = [A] RT ; p_B = [B] RT ; p_C = [C] RT ; p_D = [D] RT$$

Substituting these values in equation we have

$$K_p = \frac{[C]^l (RT)^l [D]^m (RT)^m}{[A]^j (RT)^j [B]^k (RT)^k}$$

$$K_p = \frac{[C]^l [D]^m}{[A]^j [B]^k} \times \frac{(RT)^{l+m}}{(RT)^{j+k}}$$

$$K_p = K_c \times (RT)^{(l+m) - (j+k)}$$

$$K_p = K_c \times (RT)^{\Delta n}$$

where $\Delta n = (l + m) - (j + k)$, the difference in the sums of the coefficients for the gaseous products and reactants.

From the expression it is clear that when $\Delta n = 0$, **$K_p = K_c$**

- ▶ When $\Delta n = +V_e$, the number of moles of reactant are less than the number of moles of products, thus $K_p > K_c$.
- ▶ When $\Delta n = -V_e$, the number of moles of reactant are more than the number of moles of products, thus $K_p < K_c$.
- ▶ When $\Delta n = 0$, the number of moles of reactant equals number of moles of product, thus $K_p = K_c$.

Relation Of Kp, Kc and Kx

$$K_p = \frac{(P_C)^l (P_D)^m}{(P_A)^j (P_B)^k}$$

$$K_x = \frac{[X_C]^l \times [X_D]^m}{[X_A]^j \times [X_B]^k}$$

$$K_p = \frac{[X_C P]^l \times [X_D P]^m}{[X_A P]^j \times [X_B P]^k} \frac{P^{(l+m)}}{P^{(j+k)}}$$

[Whereas, Partial Pressure = Mole fraction $\times$ total Pressure; So $p_c = X_C \cdot P$, $p_D = X_D \cdot P$, $p_A = X_A \cdot P$, $p_B = X_B \cdot P$]

$$K_p = K_c (p)^{\Delta n}$$

If $\Delta n = 0$

$$K_p = K_x$$

Now, The relation of K_p and K_c is

$$K_p = K_c (RT)^{\Delta n} \dots\dots\dots (1)$$

The relation of K_p and K_x is

$$K_p = K_x (p)^{\Delta n} \dots\dots\dots (2)$$

$$\text{And } K_p = K_x \dots\dots\dots (3)$$

Combine the equation (1) & (2)

$$K_p = K_c (RT)^{\Delta n} = K_x (p)^{\Delta n} \dots\dots\dots (4)$$

$$\text{So, } K_c (RT)^{\Delta n} = K_x (p)^{\Delta n}$$

$$K_x = K_c (RT/p)^{\Delta n}$$

$$\text{As } pV = nRT$$

$$\text{If } n=1$$

$$pV = RT$$

$$\text{Or, } V = RT/p$$

$$K_x = K_c (V)^{\Delta n}$$

$$\text{If } \Delta n = 0$$

$$\text{Then } K_x = K_c \dots\dots\dots (5)$$

From equation (3) and (5)

$$K_p = K_c = K_x$$

In the dissociation of PCl_5 as: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ if the degree of dissociation is x at equilibrium pressure P , then the the equilibrium constant for the reaction is:

	$\text{PCl}_5 (\text{g})$	$\rightleftharpoons$	$\text{PCl}_3 (\text{g})$	+	$\text{Cl}_2 (\text{g})$
Initially	a		0		0
At equilibrium	$(a-x)$		x		x
Active mass	$\frac{(a-x)}{v}$		$\frac{x}{v}$		$\frac{x}{v}$

$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{\frac{x}{v} \times \frac{x}{v}}{\frac{(a-x)}{v}} = \frac{x^2}{(a-x)v}$$

Total number of moles $= a - x + x + x = a + x$

$$P_{\text{PCl}_5} = (a - x / a + x) \times P, \quad P_{\text{PCl}_3} = (x / a + x) \times P, \quad P_{\text{Cl}_2} = (x / a + x) \times P$$

$$K_p = \frac{p_{\text{PCl}_3} \times p_{\text{Cl}_2}}{p_{\text{PCl}_5}} = \frac{\left(\frac{x}{a+x} \cdot P \right) \times \left(\frac{x}{a+x} \cdot P \right)}{\left(\frac{a-x}{a+x} \right) P} = \frac{x^2 P}{(a+x)(a-x)} = \frac{x^2 P}{a^2 - x^2}$$

Le Chatelier's principle

At the equilibrium of reversible reaction if any of the factors (Temperature, Pressure and Concentration of reactant) is changed, the equilibrium position will shift in such a way that the effect of change of factor is relieved.

There are three ways in which the stress can be caused on a chemical equilibrium:

- (1) Changing the concentration of a reactant or product.
- (2) Changing the pressure (or volume) of the system.
- (3) Changing the temperature

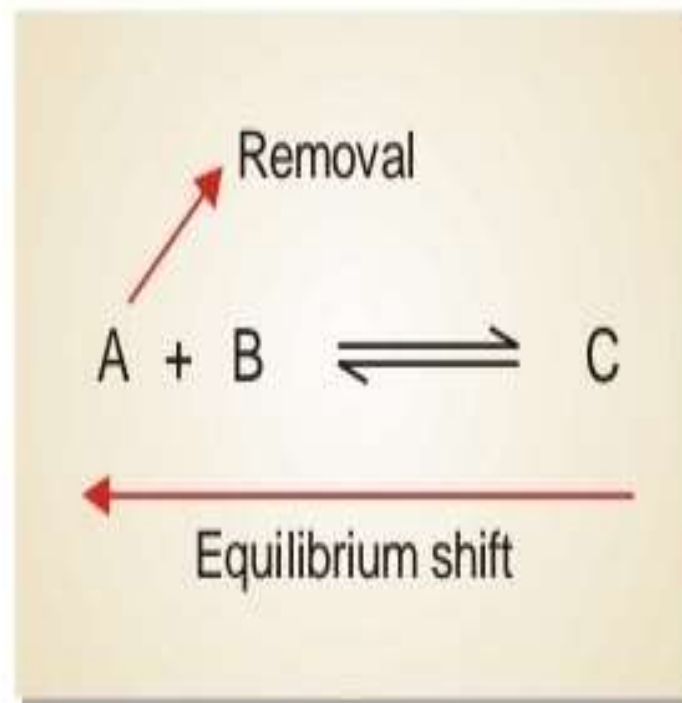
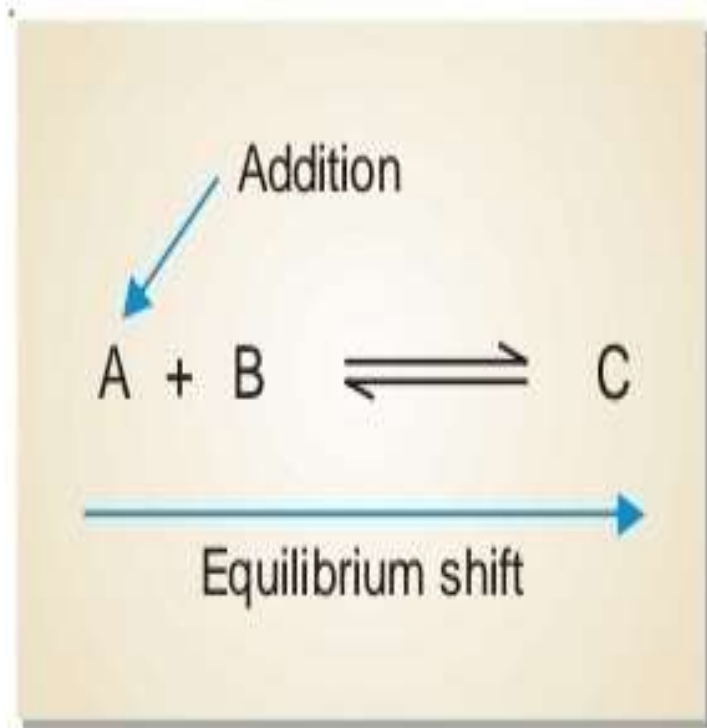
Effect Of a Change in Concentration

Let us consider a general reaction



When a reactant, say, A is added at equilibrium, its concentration is increased. The forward reaction alone occurs momentarily. According to Le Chatelier's principle, a new equilibrium will be established so as to reduce the concentration of A. Thus the addition of A causes the equilibrium to shift to right. This increases the concentration (yield) of the product C.

Following the same line of argument, a decrease in the concentration of A by its removal from the equilibrium mixture, will be undone by shift to the equilibrium position to the left. This reduces the concentration (yield) of the product C.

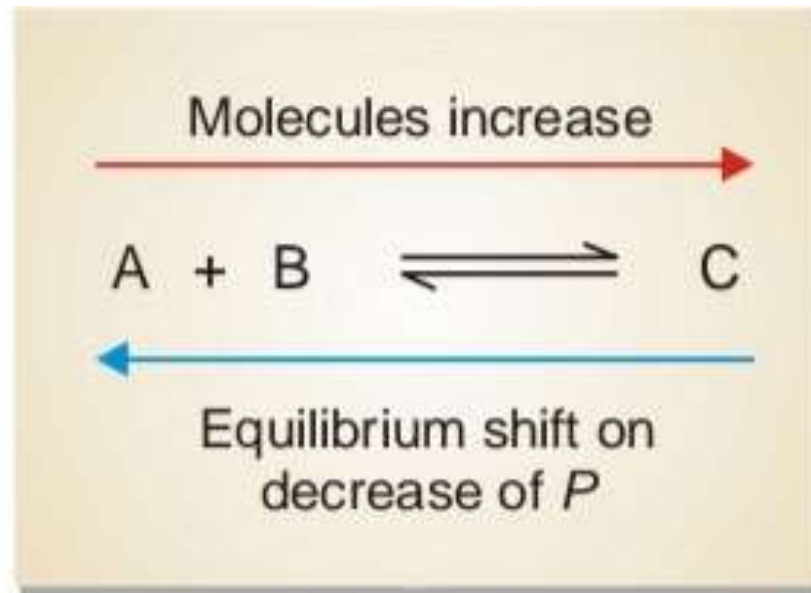
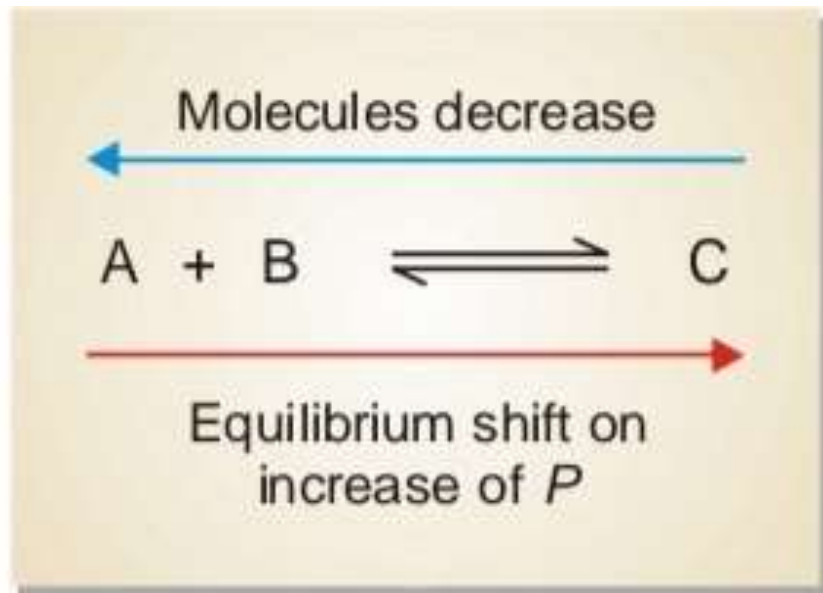


Effect Of a Change in Pressure

Let us consider a reaction,



The combination of A and B produces a decrease of number of molecules while the decomposition of C into A and B results in the increase of molecules. Therefore, by the increase of pressure on the equilibrium it will shift to right and give more C. A decrease in pressure will cause the opposite effect. The equilibrium will shift to the left when C will decompose to form more of A and B.



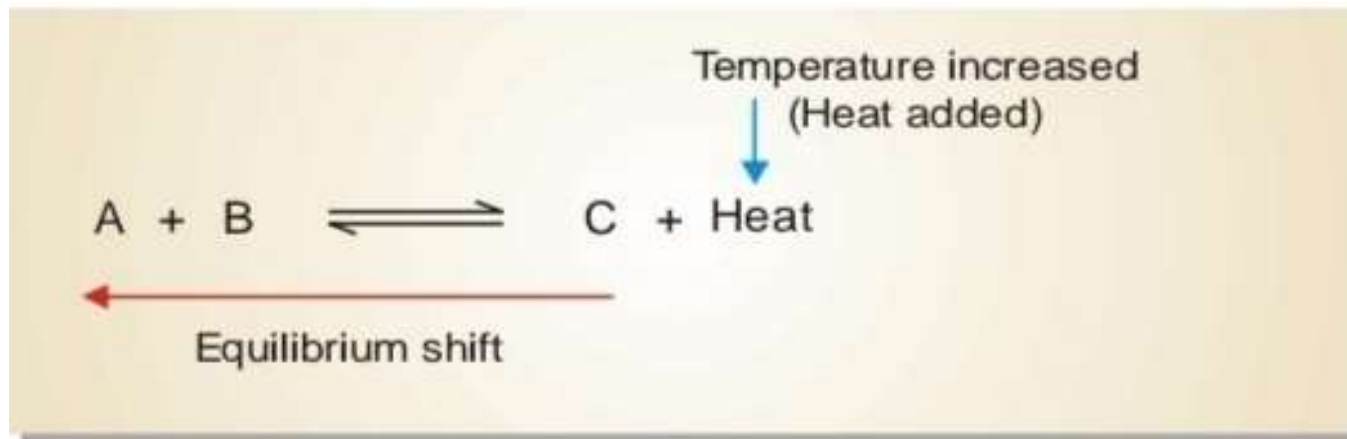
System	Pressure increased	Pressure decreased
(1) $N_2O_4 \rightleftharpoons 2NO_2$	$\leftarrow$	$\rightarrow$
(2) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$	$\leftarrow$	$\rightarrow$
(3) $2SO_2 + O_2 \rightleftharpoons 2SO_3$	$\rightarrow$	$\leftarrow$
(4) $N_2 + 3H_2 \rightleftharpoons 2NH_3$	$\rightarrow$	$\leftarrow$
(5) $H_2 + I_2 \rightleftharpoons 2HI$	No effect	No effect
(6) $N_2 + O_2 \rightleftharpoons 2NO$	No effect	No effect

Effect Of Change Of Temperature

Let us consider an **exothermic reaction**



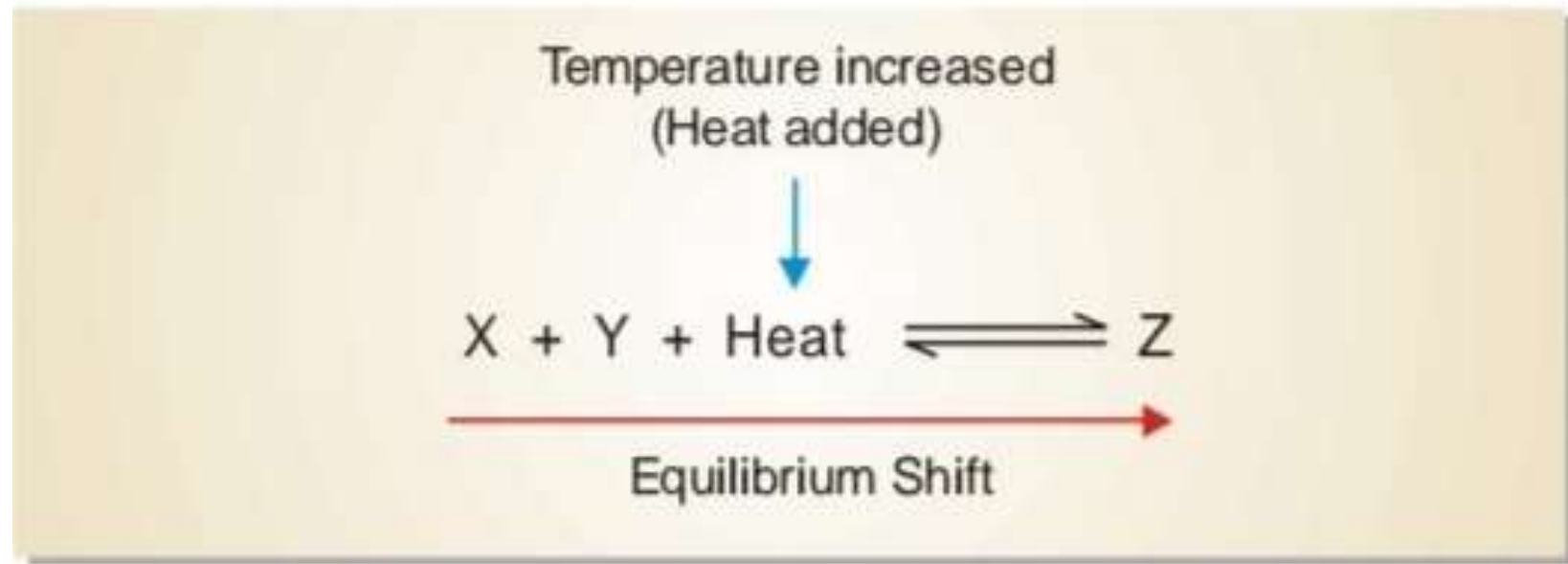
When the temperature of the system is increased, heat is supplied to it from outside. According to Le Chatelier's principle, the equilibrium will shift to the left which involves the absorption of heat. This would result in the increase of the concentration of the reactants A and B.



In an **endothermic reaction**



the increase of temperature will shift the equilibrium to the right as it involves the absorption of heat. This increases the concentration of the product.



Conditions For Maximum Yield In Industrial Processes

► Synthesis of Ammonia (Haber Process)



(1) Low temperature: By applying Le Chatelier's principle, low temperature will shift the equilibrium to the right. This gives greater yield of ammonia. In actual practice a temperature of about 450°C is used when the percentage of ammonia in the equilibrium mixture is 15.

(2) High pressure: High pressure on the reaction at equilibrium favours the shift of the equilibrium to the right. This is so because the forward reaction proceeds with a decrease in the number of moles. A pressure of about 200 atmospheres is applied in practice.

► Manufacture of Sulphuric acid (Contact Process)



(1) Low temperature: Since the forward reaction is exothermic, the equilibrium will shift on the right at low temperature. An optimum temperature between 400-450°C is required for the maximum yield of sulphur trioxide.

(2) High pressure: Since the number of moles are decreased in the forward reaction, increase of pressure will shift the equilibrium to the right. Thus for maximum yield of SO₃, 2 to 3 atmosphere pressure is used.

► Manufacture of Nitric acid (Birkeland-Eyde process)

Nitric acid is prepared on a large scale by making use of the



(1) High temperature: Since the forward reaction is endothermic, increase of temperature will favour it (Le Chatelier's principle). Thus a high temperature of the order of 3000°C is employed to get high yield of nitric acid.

(2) No effect of pressure: Since the forward reaction involves no change in the number of moles, a change in pressure has no effect on the equilibrium.

(3) High concentration: The formation of nitric oxide is favoured by using high concentrations of the reactants i.e. N_2 and O_2